

SUPSI

University of Applied Sciences and Arts of Southern Switzerland

Designing With

A New Educational Module to Integrate
Artificial Intelligence, Machine Learning and
Data Visualization in Design Curricula

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Didactic Guidelines

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3. Module 1: Getting Familiar

This module teaches students to employ AI to train, generate, analyse, and visualise data for specific design purposes {Module 1-Obj} (Fig. 6). The module is organised into three sub-modules to get familiar with a specific discipline: Machine Learning (ML), Generative AI (Gen AI) and Data Visualisation (DV). Each sub-module is intended to provide basic literacy, guided tutorials of tools, hands-on experience and documentation practices of results. To promote the development of personal skills, students are suggested to work individually.

Teachers can arrange the duration of each activity based on the level of in-depth study required. Sub-modules are not strictly related to each other. The application of all of them is not mandatory, teachers can make a selection based on needs and interests. The structure of the sub-modules is not designed based on the tool mentioned in the guideline. For a more complete list of tools, refer to the Designing With framework [\[link\]](#).

Educational Objectives

{MOD1-Obj}

Students will learn to

employ AI to train, generate, analyse and visualise data

for specific design purposes

Instructional Activities

{MOD1-Act.1}

Activity intended to

provide students

with vocabulary knowledge of AI

{MOD1-Act.2}

Activity intended to

provide students

with basic functionality of AI

{MOD1-Act.3}

Activity intended to

provide students

with practical examples of AI applied to design

{MOD1-Act.4}

Activity intended to

explain

the procedure of train, generate, analyse and visualise data with AI

{MOD1-Act.4.1}

Activity intended to

introduce students

to familiarising, collecting or creating data

{MOD1-Act.4.2}

Activity intended to

set or align

with a design goal

{MOD1-Act.4.3}

Activity intended to

provide students

a step-by-step tool tutorial, such as RunwayML, Dall-E, Vision API

{MOD1-Act.5}

Activity intended to

allow students

to individually train, generate, analyse and visualise data with AI

{MOD1-Act.6}

Activity intended to

provide students

with a structured template for process and results documentation

{MOD1-Act.7}

Activity intended to

make students

explaining and questioning the process

Module 1: Getting Familiar	Cognitive process dimension					
	(1) Remember	(2) Understand	(3) Apply	(4) Analyze	(5) Evaluate	(6) Create
	(a) Factual Knowledge		{MOD1-Act.1}			
	(b) Conceptual Knowledge		{MOD1-Act.2} {MOD1-Act.3}			
	(c) Procedural Knowledge		{MOD1-Act.4} {MOD1-Act.4.1, 4.2, 4.3}	{MOD1-Act.5} {MOD1-Obj}		
	(d) Metacognitive Knowledge				{MOD1-Act.6}	{MOD1-Act.7}

Fig. 6 Module 1: 'Getting Familiar'. Educational Objectives and Instructional Activities according to the Bloom Taxonomy Table.

3.1 Sub-module 1: Getting familiar with Machine Learning (ML)

This sub-module teaches students to train simplified ML models to create design artefacts {ML-Obj}. Students develop procedural knowledge through practical guided activities (e.g. RunwayML step-by-step tool tutorials) and they are asked to apply the acquired procedure to a familiar task (e.g. creating a logo). Students are first provided with a basic literacy of ML including technical vocabulary {ML-Act1} and primary functionalities {ML-Act2}. This activity is meant to promote factual and conceptual knowledge by teaching the basic elements of the discipline and their interrelationship to explain the functionality of ML tools. Once basic literacy is introduced, specific case studies are provided to contextualise the application of ML within the design practice {ML-Act3}. At this point, the teaching moves from a theoretical to a procedural level. In this direction,

students are first introduced to a guided tutorial {ML-Act4.3} on how to use an ML tool (e.g. RunwayML) starting from setting a design goal {ML-Act4.1} and creating their dataset {ML-Act4.2}. Then, to familiarise themselves with the procedure, students individually apply what they learned by training an ML model aligned with the design goal {ML-Act5}. Lastly, moving the focus to meta-cognitive knowledge, students are asked to analyse and document the process by breaking down the steps and selecting the relevant results {ML-Act6}. This activity aims to create awareness and knowledge of cognition. To foster participation and sharing of results, students are asked to present their work and question others {ML-Act7} (Fig. 7).

Educational Objectives

{ML-Obj}

Students will learn to

train a Machine Learning model

for a specific design purpose

Instructional Activities

{ML-Act.1}

Activity intended to

provide students

with vocabulary knowledge of ML

{ML-Act.2}

Activity intended to

provide students

with basic functionality of ML

{ML-Act.3}

Activity intended to

provide students

with practical examples of ML applied to design

{ML-Act.4}

Activity intended to

explain

training a simplified ML model

{ML-Act.4.1}

Activity intended to

set or align

with a design goal

{ML-Act.4.2}

Activity intended to

introduce students

to familiarising, collecting or creating data

{ML-Act.4.3}

Activity intended to

provide students

a step-by-step tool tutorial, such as RunwayML

{ML-Act.5}

Activity intended to

allow students

to individually train a simplified MLmodel for a design goal

{ML-Act.6}

Activity intended to

provide students

with a structured template for process and results documentation

{ML-Act.7}

Activity intended to

make students

explaining and questioning the process

Sub-module 1: Getting familiar with ML	Cognitive process dimension					
	(1) Remember	(2) Understand	(3) Apply	(4) Analyze	(5) Evaluate	(6) Create
(a) Factual Knowledge		{ML-Act.1}				
(b) Conceptual Knowledge		{ML-Act.2} {ML-Act.3}				
(c) Procedural Knowledge		{ML-Act.4} {ML-Act.4.1, 4.2, 4.3}	{ML-Act.5} {ML-Obj}			
(d) Metacognitive Knowledge				{ML-Act.6}	{ML-Act.7}	

Fig. 7 Sub-Module 1: ‘Getting Familiar’ with Machine Learning. Educational Objectives and Instructional Activities arranged into the Bloom Taxonomy Table.

3.2 Sub-module 2: Getting familiar with Generative AI (genAI)

Students are first provided with a basic literacy of AI including technical vocabulary {GenAI-Act1} and primary functionalities {GenAI-Act2}. This activity is meant to promote factual and conceptual knowledge by teaching the basic elements of the discipline and their interrelationship to explain the functionality of AI tools. Once basic literacy is introduced, specific case studies are provided to contextualise the application of AI within the design practice {GenAI-Act3}. At this point, the teaching moves from a theoretical to a procedural level. In this direction, students are first introduced to a guided tutorial {GenAI-Act4.3} on how to use a Generative AI tool (e.g. Midjourney) starting from setting a design goal

{GenAI-Act4.1} and experimenting with relative prompts {GenAI-Act4.2} Then, to familiarise themselves with the procedure, students individually apply what they learned by refining selected prompts and generating content based on the design goal {GenAI-Act5}. Lastly, moving the focus to meta-cognitive knowledge, students are asked to analyse and document the process by breaking down the steps and selecting the relevant results {GenAI-Act6}. This activity aims to create awareness and knowledge of cognition. To foster participation and sharing of results, students are asked to present their work and question others {GenAI-Act7}.

Educational Objectives

{GenAI-Obj}

Students will learn to

generate contents with GenAI tools

to create design artifacts

Instructional Activities

{GenAI-Act.1}

Activity intended to

provide students

with vocabulary knowledge of GenAI

{GenAI-Act.2}

Activity intended to

provide students

with basic functionality of GenAI

{GenAI-Act.3}

Activity intended to

provide students

with practical examples of GenAI applied to design

{GenAI-Act.4}

Activity intended to

explain

the procedure of generating contents with GenAI

{GenAI-Act.4.1}

Activity intended to

set or align

with a design goal

{GenAI-Act.4.2}

Activity intended to

introduce students

to familiarising, collecting or creating prompts (data)

{GenAI-Act.4.3}

Activity intended to

provide students

a step-by-step tool tutorial, such as Dall-E or Midjoruney

{GenAI-Act.5}

Activity intended to

allow students

to individually generate contents with GenAI for a design goal

{GenAI-Act.6}

Activity intended to

provide students

with a structured template for process and results documentation

{GenAI-Act.7}

Activity intended to

make students

explaining and questioning the process

Sub-module 2: Getting familiar with GenAI	Cognitive process dimension					
	(1) Remember	(2) Understand	(3) Apply	(4) Analyze	(5) Evaluate	(6) Create
	(a) Factual Knowledge		{GenAI-Act.1}			
	(b) Conceptual Knowledge		{GenAI-Act.2} {GenAI-Act.3}			
	(c) Procedural Knowledge		{GenAI-Act.4} {GenAI-Act.4.1, 4.2, 4.3}	{GenAI-Act.5} {GenAI-Obj}		
	(d) Metacognitive Knowledge				{GenAI-Act.6}	{GenAI-Act.7}

Fig. 8 Sub-Module 1: 'Getting Familiar' with Generative AI. Educational Objectives and Instructional Activities arranged into the Bloom Taxonomy Table.

3.3 Sub-module 2: Getting familiar with Data Visualisation (DV)

This sub-module teaches students to apply Computer Vision (CV) algorithms to analyse and visualise data {DV-Obj}. Students develop procedural knowledge through practical guided activities and they are asked to apply the acquired procedure to a familiar task (e.g. creating an infographic). Students are first provided with a basic literacy of CV algorithms and DV techniques including technical vocabulary {DV-Act1} and primary functionalities {DV-Act2}. This activity is meant to promote factual and conceptual knowledge by teaching the basic elements of the disciplines (CV and DV) and their interrelationship in analysing and visualising complex phenomena (Omena, 2021). Once basic literacy is introduced, specific case studies are provided to contextualise the application of CV within the design research {DV-Act3}. At this point, the teaching moves from a theoretical to a procedural level.

In this direction, students are first introduced to a guided tutorial {DV-Act4.3} on how to analyse data with CV algorithms (e.g. Meme Spector) and visualise the results (e.g. Gephi) starting from setting a research question (DV-Act4.1) and collecting and organising data related to the phenomena {DV-Act4.2}. Then, to familiarise themselves with the procedure, students individually apply what they learned by investigating the dataset to answer the research question {DV-Act5}. Lastly, moving the focus to meta-cognitive knowledge, students are asked to document the process by breaking down the steps and selecting the relevant results {DV-Act6}. This activity aims to create awareness and knowledge of cognition. To foster participation and sharing of results, students are asked to present their work and question others {DV-Act7} (Fig. 9).

Educational Objectives

{DV-Obj}

Students will learn to

apply Computer Vision algorithms

for analyse and visualise data

Instructional Activities

{DV-Act.1}

Activity intended to

provide students

with vocabulary knowledge of CV and DV

{DV-Act.2}

Activity intended to

provide students

with basic functionality of CV algorithms and DV techniques

{DV-Act.3}

Activity intended to

provide students

with practical examples of CV applied to design research

{DV-Act.4}

Activity intended to

explain

the procedure of analysing data with CV algorithms

{DV-Act.4.1}

Activity intended to

introduce students

to familiarising, collecting or creating prompts (data)

{DV-Act.4.2}

Activity intended to

set or align

with a research question

{DV-Act.4.3}

Activity intended to

provide students

a step-by-step tool tutorial, such as Memespector, Gephi

{DV-Act.5}

Activity intended to

allow students

to individually analyse data with computer vision for a research goal

{DV-Act.6}

Activity intended to

provide students

with a structured template for process and results documentation

{DV-Act.7}

Activity intended to

make students

explaining and questioning the process

Sub-module 3: Getting familiar with DV	Cognitive process dimension					
	(1) Remember	(2) Understand	(3) Apply	(4) Analyze	(5) Evaluate	(6) Create
	(a) Factual Knowledge	{DV-Act.1}				
	(b) Conceptual Knowledge	{DV-Act.2} {DV-Act.3}				
	(c) Procedural Knowledge	{DV-Act.4} {DV-Act.4.1, 4.2, 4.3}	{DV-Act.5} {DV-Obj}			
(d) Metacognitive Knowledge				{DV-Act.6}	{DV-Act.7}	

Fig. 9 Sub-Module 1: ‘Getting Familiar’ with Data Visualisation. Educational Objectives and Instructional Activities arranged into the Bloom Taxonomy Table.

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